

```
from google.colab import files
uploaded = files.upload()
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Saving Titanic-Dataset.csv to Titanic-Dataset.csv

```
import pandas as pd
```

```
df = pd.read_csv('Titanic-Dataset.csv') # replace with your
df.head()
```

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh
1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0
3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0
5	0	3	Allen, Mr. William Henry	male	35.0	0	0

```
import pandas as pd
import matplotlib.pyplot as plt
```

```

import seaborn as sns

# Load Dataset
df = pd.read_csv('/content/Titanic-Dataset.csv')

# Dataset Information
print(df.info())

# Missing Values
print(df.isnull().sum())

# Survival Count
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()

# Survival by Gender
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Survival by Gender")
plt.show()

# Survival by Passenger Class
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title("Survival by Passenger Class")
plt.show()

# Age Distribution
sns.histplot(df['Age'].dropna(), bins=20)
plt.title("Age Distribution")
plt.show()

# Correlation Heatmap
numeric_df = df.select_dtypes(include=['number'])
sns.heatmap(numeric_df.corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()

```



```

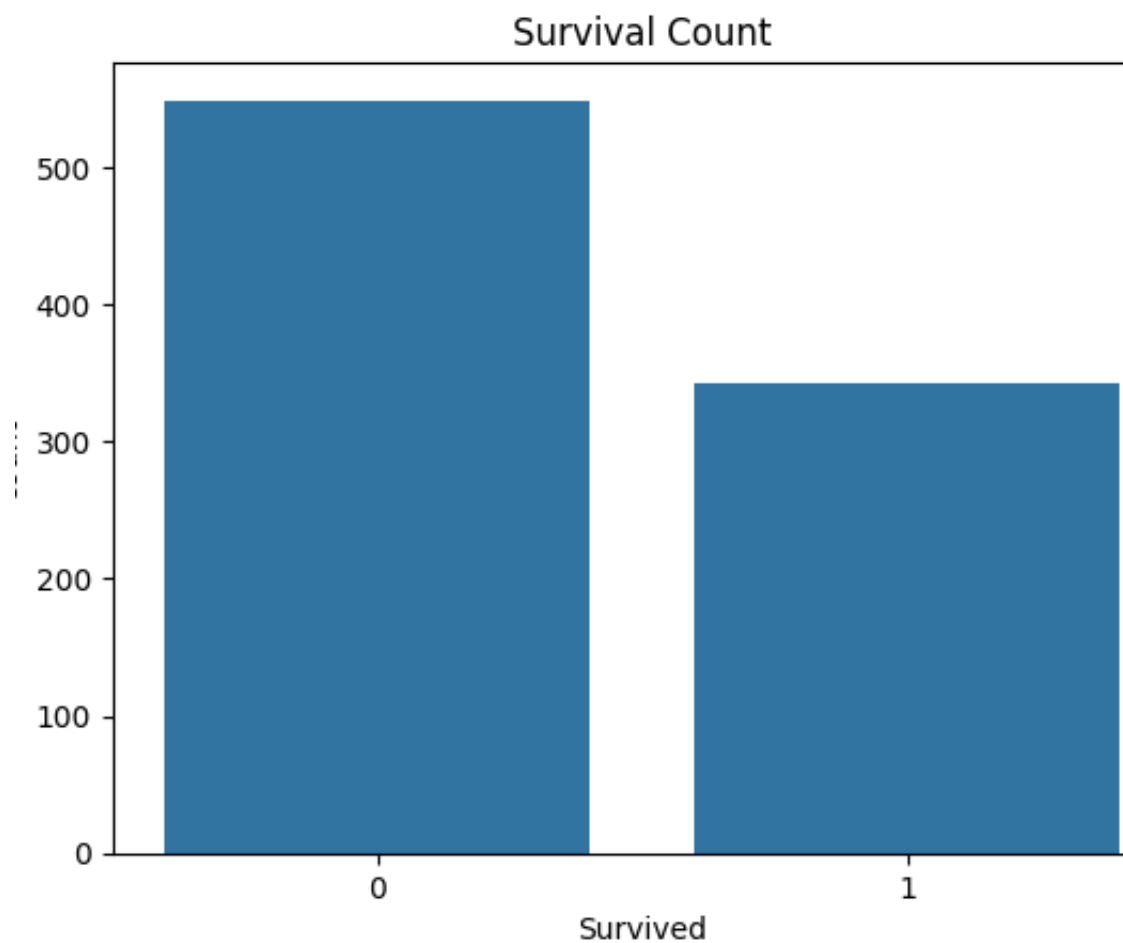
class 'pandas.core.frame.DataFrame'>
ageIndex: 891 entries, 0 to 890
ta columns (total 12 columns):
  Column      Non-Null Count  Dtype
-  -----  -
  PassengerId  891 non-null    int64
  Survived     891 non-null    int64
  Pclass       891 non-null    int64
  Name         891 non-null    object
  Sex          891 non-null    object

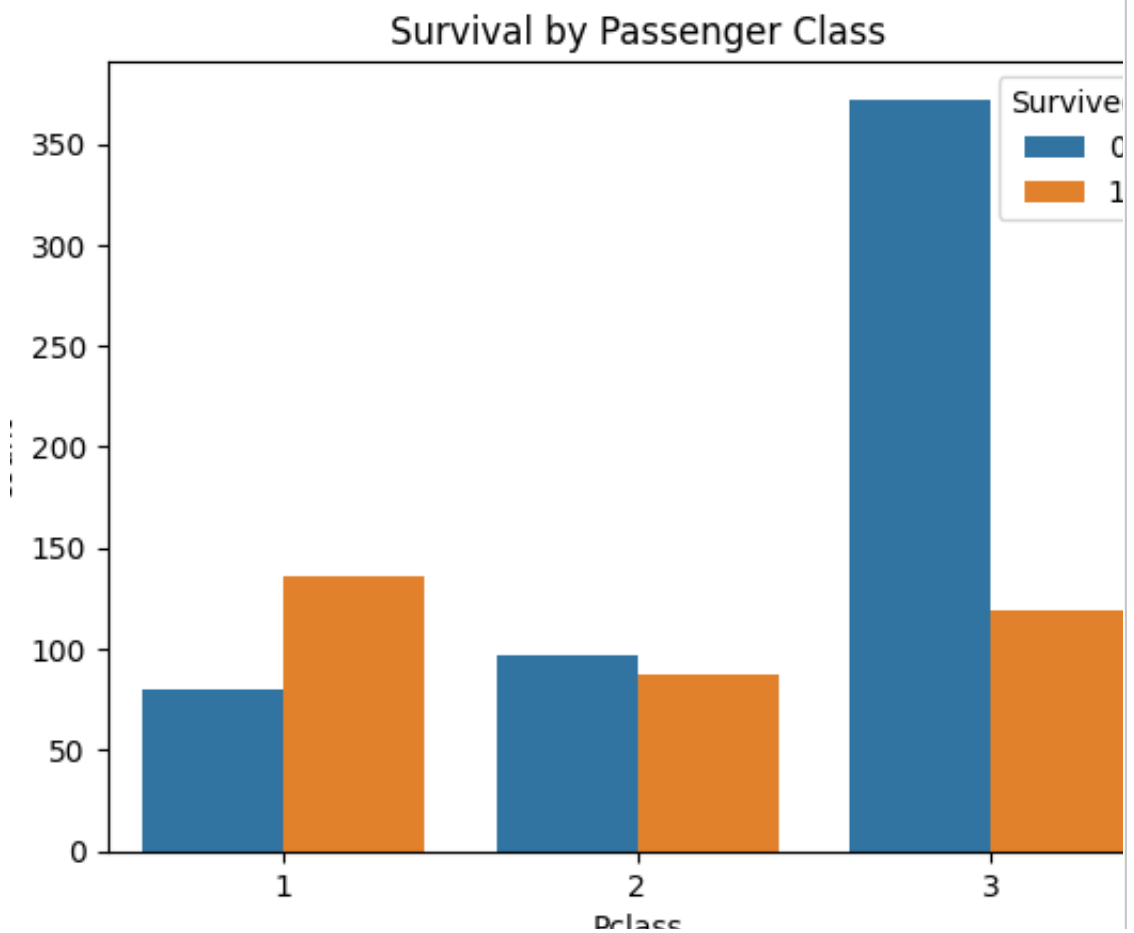
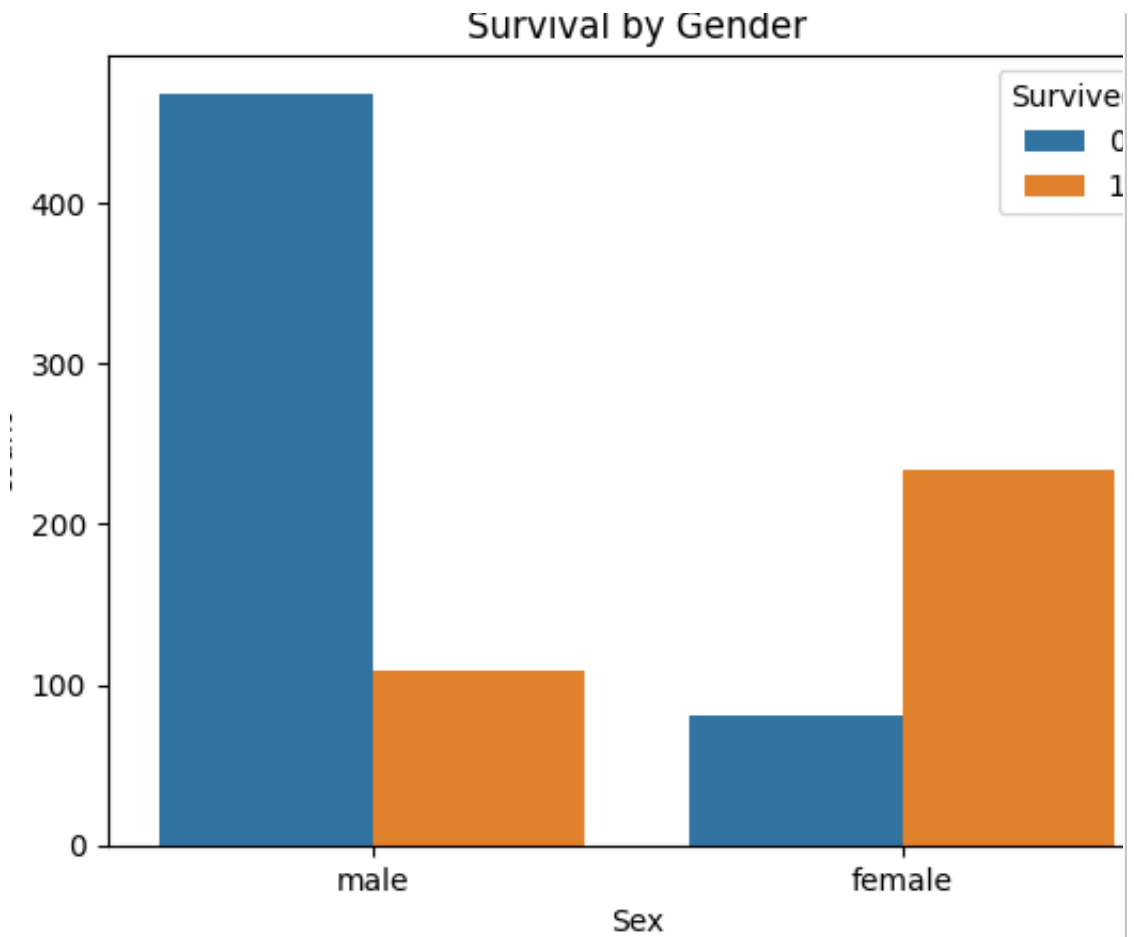
```

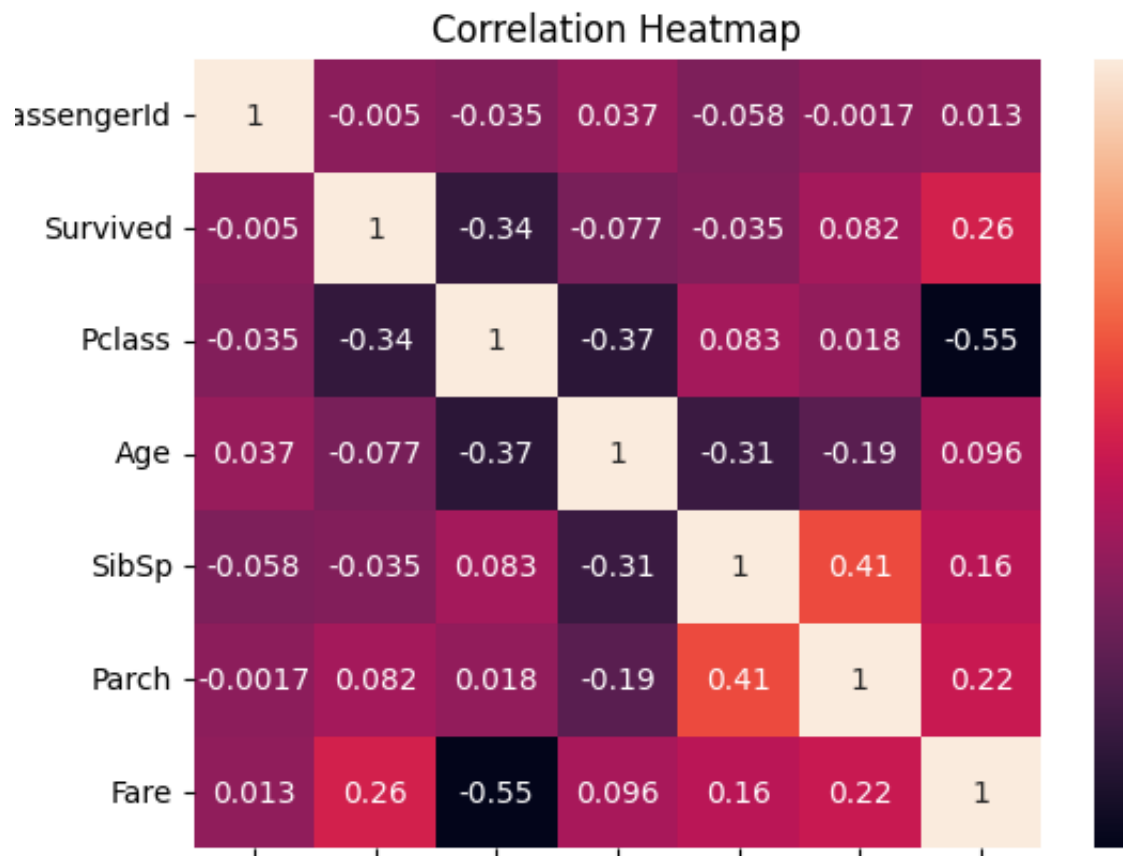
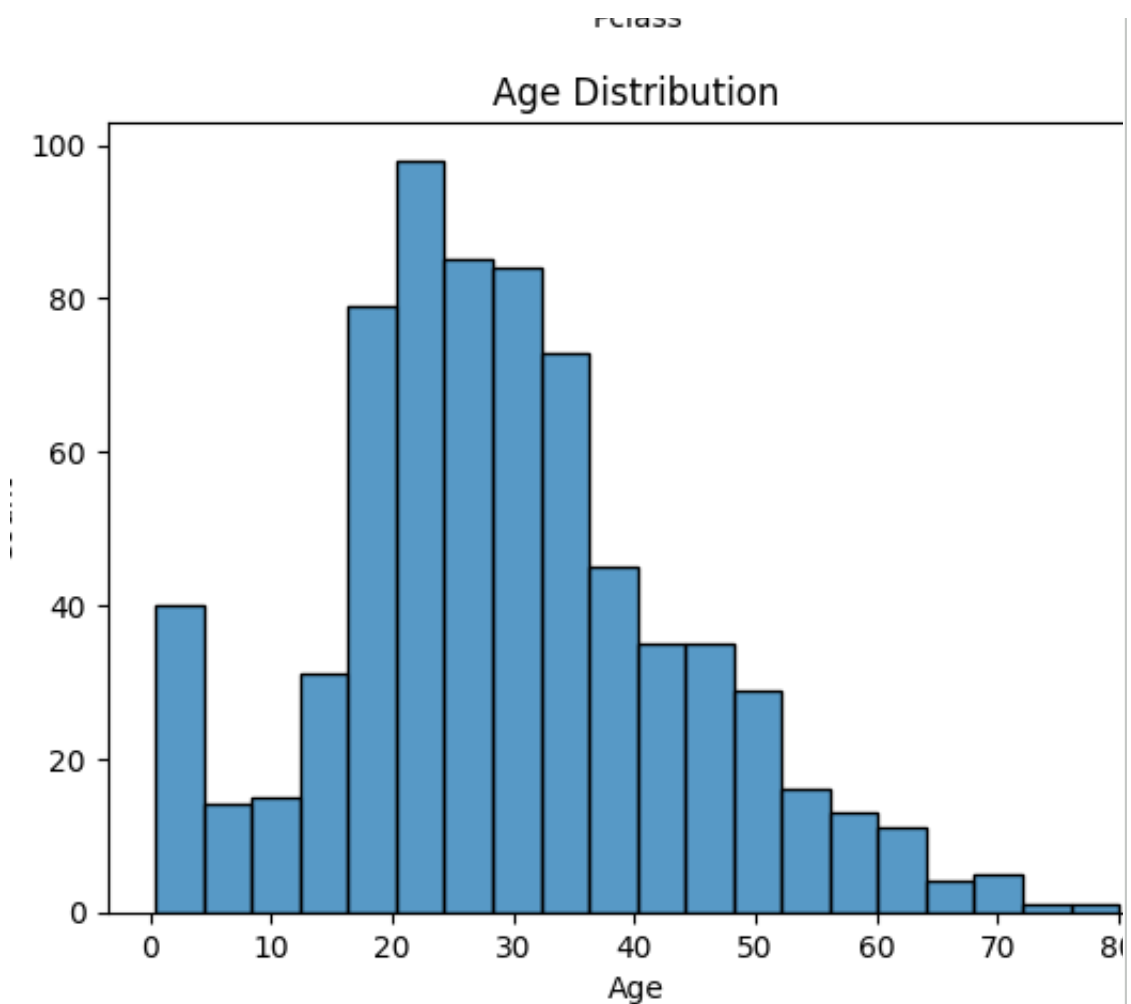
```

Age          714 non-null    float64
SibSp        891 non-null    int64
Parch        891 non-null    int64
Ticket       891 non-null    object
Fare         891 non-null    float64
0 Cabin      204 non-null    object
1 Embarked   889 non-null    object
types: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
PassengerId  0
Survived      0
Class        0
Name         0
Sex          0
Age          177
SibSp        0
Parch        0
Ticket       0
Fare         0
Cabin        687
Embarked     2
type: int64

```







PassengerId

Survived

Pclass

Age

SibSp

Parch

Fare